

# Statistics

GCSE Maths · Practice Questions

20 questions · 62 marks total

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## Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

**1** [1 mark]

Write down the mode of these numbers.

**3 7 4 7 2 9 7**

.....

**2** [2 marks]

Work out the mean of these numbers.

**4 8 5 11 7**

.....

**3** [2 marks]

Here are seven numbers.

**12 5 9 15 3 9 20**

(a) Work out the median.

(b) Work out the range.

.....

**4** [2 marks]

The table shows the favourite sport of some students.

| Sport     | Football | Netball | Rugby | Tennis |
|-----------|----------|---------|-------|--------|
| Frequency | 12       | 8       | 5     | 7      |

(a) Write down the modal sport.

(b) How many students were asked altogether?

.....

5

[3 marks]

The table shows the number of pets owned by 25 families.

| Number of pets | 0 | 1  | 2 | 3 |
|----------------|---|----|---|---|
| Frequency      | 8 | 10 | 5 | 2 |

Work out the mean number of pets per family.

.....

6

[3 marks]

Describe the type of correlation you would expect to see on a scatter graph of

- (a) the age of a car and its value
- (b) the height of a person and their shoe size
- (c) the house number a person lives at and their shoe size.

.....

7

[3 marks]

90 people were asked to name their favourite colour. The results are shown in a pie chart.

- (a) 30 people chose blue. Work out the angle of the blue sector.
- (b) The angle of the red sector is  $72^\circ$ . How many people chose red?

.....

8

[3 marks]

The table shows the time,  $t$  minutes, that 30 people waited for a bus.

| Time ( $t$ minutes) | $0 < t \leq 10$ | $10 < t \leq 20$ | $20 < t \leq 30$ | $30 < t \leq 40$ |
|---------------------|-----------------|------------------|------------------|------------------|
| Frequency           | 5               | 12               | 8                | 5                |

- (a) Write down the modal class.
- (b) Write down the class interval that contains the median.

.....

9

[4 marks]

Use the same table of waiting times.

| Time (t minutes) | $0 < t \leq 10$ | $10 < t \leq 20$ | $20 < t \leq 30$ | $30 < t \leq 40$ |
|------------------|-----------------|------------------|------------------|------------------|
| Frequency        | 5               | 12               | 8                | 5                |

(a) Work out an estimate for the mean waiting time.

(b) Explain why your answer is only an estimate.

10

[4 marks]

Class A has 10 students and their mean test score is 62.

Class B has 15 students and their mean test score is 68.

(a) Work out the mean score of all 25 students.

(b) The range for class A is 30 and the range for class B is 12. Write down one comparison you can make between the two classes.

## Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

The mean of eight numbers is 15.

One of the numbers, 22, is removed.

Work out the mean of the remaining seven numbers.

2

[2 marks]

Ella wants to find out how much exercise adults in her town do each week.

She asks 50 people as they leave a gym.

Give one reason why her sample may be biased, and suggest one improvement.

- 3

[3 marks]

A school has 300 students in Year 10 and 200 students in Year 11.  
A stratified sample of 75 students is taken from these two year groups.  
Work out how many students should be taken from each year group.

- 4

[3 marks]

The cumulative frequency table shows the heights,  $h$  cm, of 80 plants.

| Height ( $h$ cm)     | $h \leq 150$ | $h \leq 160$ | $h \leq 170$ | $h \leq 180$ | $h \leq 190$ |
|----------------------|--------------|--------------|--------------|--------------|--------------|
| Cumulative frequency | 8            | 24           | 56           | 72           | 80           |

Use the table to estimate the median height.

- 5

[4 marks]

The table shows the times,  $t$  minutes, taken by 87 people to finish a puzzle.

| Time ( $t$ minutes) | $0 < t \leq 10$ | $10 < t \leq 20$ | $20 < t \leq 40$ | $40 < t \leq 70$ |
|---------------------|-----------------|------------------|------------------|------------------|
| Frequency           | 15              | 24               | 30               | 18               |

(a) Work out the frequency density for each class.  
(b) Estimate the number of people who took between 30 and 40 minutes.

- 6

[4 marks]

Box plots were drawn for the test marks of a group of boys and a group of girls.

|       | Min | LQ | Median | UQ | Max |
|-------|-----|----|--------|----|-----|
| Boys  | 12  | 20 | 26     | 33 | 45  |
| Girls | 15  | 24 | 28     | 32 | 40  |

(a) Work out the interquartile range for each group.  
(b) Make two comparisons between the marks of the boys and the girls.

7

[4 marks]

A scatter graph was drawn of revision time,  $x$  hours, against test score,  $y$  marks, for 12 students who each revised between 2 and 12 hours.

The line of best fit has equation  $y = 2.5x + 12$

- (a) Estimate the score of a student who revised for 8 hours.
  - (b) Interpret the number 2.5 in the context of the question.
  - (c) Explain why the line should not be used to estimate the score of a student who revised for 40 hours.
- 

8

[4 marks]

The table shows the weights,  $w$  kg, of 40 parcels.

| Weight ( $w$ kg) | $0 < w \leq 20$ | $20 < w \leq 40$ | $40 < w \leq 60$ | $60 < w \leq 80$ |
|------------------|-----------------|------------------|------------------|------------------|
| Frequency        | 7               | 13               | 15               | 5                |

Work out an estimate for the mean weight of a parcel.

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9

[4 marks]

A group contains 20 people.

The mean age of the 12 women in the group is 34 years.

The mean age of all 20 people is 31 years.

Work out the mean age of the 8 men.

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A histogram was drawn for the distances,  $d$  km, travelled by 82 delivery drivers. The bars have these frequency densities.

| Distance ( $d$ km) | $0 < d \leq 10$ | $10 < d \leq 20$ | $20 < d \leq 50$ | $50 < d \leq 80$ |
|--------------------|-----------------|------------------|------------------|------------------|
| Frequency density  | 1.2             | 2.8              | 1.0              | 0.4              |

- (a) Work out the frequency of each class and show that the total is 82.
- (b) Estimate the number of drivers who travelled more than 40 km.
- (c) Estimate the median distance.